MIL-F-50542A (AR)
14 June 1978
SUPERSEDING
MIL-F-50542 (MU)
15 July 1970

MILITARY SPECIFICATION

FUZE FOR FLECHETTE WARHEAD, WDU-4A/A (WITH EXPULSION,CHARGE)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts and assembly of a Fuze for Flechette Warhead, WDU-4A/A (with Expulsion Charge).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Specification For.
MIL-P-60412(MU)	- Packaging, Packing and Marking For Shipment of Artillery Type and Rocket Fuzes, General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1340

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DRAWINGS

USAF

65C11046 - Fuze for Flechette Warhead, WDU-4A/A (with Expulsion Charge)
67D9700 - Warhead 2.75 Inch Rocket, Flechette WDU-4A/A

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable specifications and drawings.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawings 65C11046 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Fuze Assembly Functioning. - The Expulsion Charge Assembly shall completely expel the warhead cargo upon firing of the fuze assembly.

3.4 First Article Inspection. - This specification makes provisions for first article inspection. Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

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3.5 Data Requirements. - The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.5.4.

3.6 Workmanship .- All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, corrosion products and other foreign matter. The assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of the industry. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Fuze for Flechette Warhead, WDU-4A/A (with Expulsion Charge)	65C11046	50
Charge Assembly, Expulsion	65C1105O	25

4.3.2 Inspection to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-50542A (AR)

PARAGRAPH	TITLE: Fuze for Flechette Warhead WDU-4A/A (With Expulsion Charge)		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze for Flechette Warhead, WDU-4A/A (With Expulsion Charge)				
	Dwg 65C11046				
	Examination for Defects	50		3.2	4.3.2.2
	Fuze Assembly Sealing Test	50		3.2	4.5.2
	Fuze Assembly Functional Test	50		3.3	4.5.3
	Charge Assembly, Expulsion				
	Dwg 65C11050				
	Examination for Defects	25		3.2	4.3.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. -See MIL-A-48078

a. Sampling plans. Unless otherwise specified in the Classifications of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 65C11050
4.3.2.1	Charge Assembly, Expulsion				NEXT HIGHER ASSEMBLY 65C11046
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Thickness of Assembly		0.40%	3.2	Gage
102	Weight		0.40%	3.2	Gage
103	Charge Assembly Improperly Sealed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.2	Fuze for Flechette Warhead, WDU-4A/A (With Expulsion Charge)				65C11046
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Overall length, max		0.40%	3.2	Gage
102	Fuze case lid not flush with Fuze assembly		0.40%	3.2	Visual
103	Components missing or broken		0.40%	3.2	Visual
104	Fuze assembly seal test		0.40%	3.2	4.5.2
105	Fuze, assembly functioning	50		3.3	4.5.3
Minor					
201	Marking missing or incomplete		0.65%	3.2	Visual
202	Foreign material inside of fuze case		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4 .4.3 Testing .-Testing is described in the First Article and Quallty Conformance Inspection tables.

4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and bound to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I. Accept-reject criteria shall be in accordance with MIL-STD-105.

4.5.2 Fuze assembly sealing. The fuze assembly shall be subjected to the seal leak test as specified on dwg 65C11046. Any fuze assembly not meeting test requirements shall be classed defective.

4.5.3 Fuze assembly functioning test. - The fuze assembly mounted in a simulated warhead body shall be accelerated on a centrifuge to 35.0 gravities (g's) plus or minus one (1) g for a minimum of 1.5 seconds. The fuze will normally fire upon a decrease in acceleration to 15.0 g's or below after arming at 35.0 g's. After deceleration, fuzes which have not fired shall be examined for the safe position before handling. A fuze which has armed but has not fired shall be handled using an approved safety method to prevent injury. If one or more fuze assemblies fail to function or fail to shear the simulated warhead pins, the lot shall be rejected.

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4.5.4 Data submission. - The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with Data Item Descriptions DI-R-1721 and DI-R1724 giving the detailed test results.

5. PACKAGING

5.1 Packaging and marking. -Level C - The fuze, less expulsion charge, shall be packaged in accordance with MIL-P-60412. Marking shall be in accordance with drawing requirements.

5.2 Packing and marking. -Level C - The packaged fuzes shall be packed and marked in accordance with MIL-P-60412. The ICC nomenclature shall be "Percussion Fuzes."

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use in the 2.75 Inch Rocket System.

6.2 Ordering data. -See MIL-A-48078

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-1, Dover, NJ 07801.

6.4 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph	Data Requirement	Applicable (DI D)
4.5.4	Quality Inspection Defect Report	DI-R-1721
4.5.4	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724

(Copies of data item descriptive required by the contractor in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer)

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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